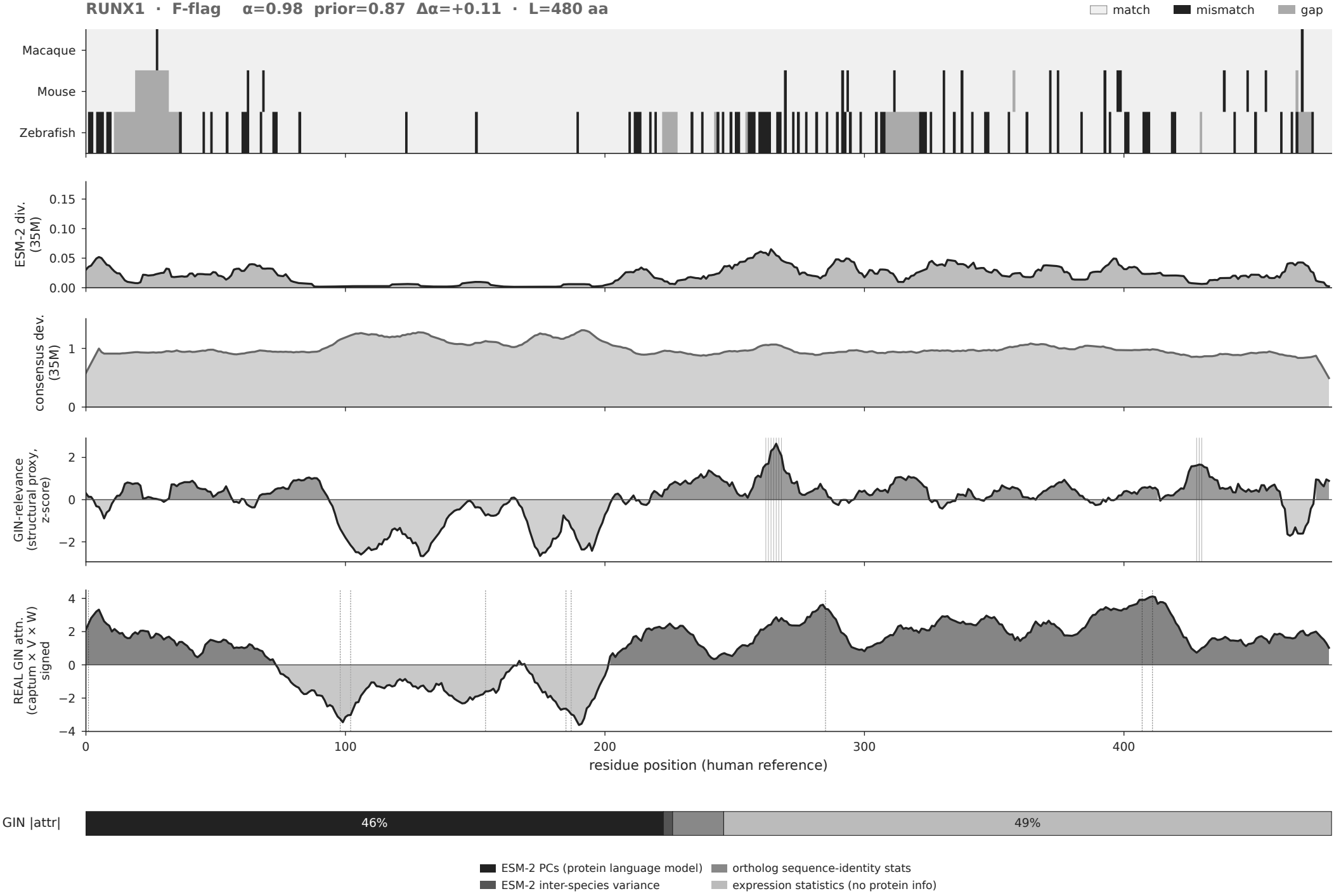

RUNX1 · F-flag α=0.98 prior=0.87 Δα=+0.11 · L=480 aa
match
mismatch
gap
Macaque
Mouse
Zebrafish
ESM-2 div. (35M)
0.15
0.10
0.05
0.00
consensus dev. (35M)
1
0
GIN-relevance (structural proxy, z-score)
2
0
−2
REAL GIN attn. (captum × V × W) signed
4
2
0
−2
−4
0
100
200
300
400
residue position (human reference)
GIN |attr|
46%
49%
ESM-2 PCs (protein language model)
ESM-2 inter-species variance
ortholog sequence-identity stats
expression statistics (no protein info)